

Name:

Date:

Period:

HONORS PHYSICS FALL FINAL EXAM

(Version A)



Please do not open this packet until instructed to do so!

Rules

1. **Phones, earbuds, headphones, smartwatches, computers, and other electronics must be put away in your backpacks against the wall during the entirety of the exam. If electronics are out for *any reason*, you will receive a zero and a detention!**
2. Put your name three places: On your Equation Sheet (back of this page), On your Scratch Work sheet (next page), and on the first page of the Exam Packet.
3. Bubble in the Key Version (A) on your Scantron
4. When you are finished, turn in your scantron, equation sheet, and exam packet up front.
5. Stay silent until all your classmates are finished. Electronics (especially phones, but also earbuds, headphones, laptops, smartwatches, etc.) need to be kept away until the instructor indicates that everyone is finished.

Name:

Date:

Period:

Equation Sheet

Basic Kinematics

$$\bar{v} \equiv \frac{\Delta x}{\Delta t}$$

$$\bar{a} \equiv \frac{\Delta v}{\Delta t}$$

$$\bar{v} = \frac{v + v_0}{2}$$

Kinematic Equations

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

“Old Faithful”

“Big Chalupa”

“Ain’t Got No Time”

Range Equation

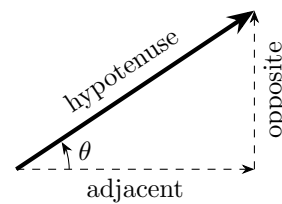
$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Vectors & Trig

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$



Forces

$$\Sigma F = ma$$

$$w = mg$$

$$f_s = \mu_s N$$

$$f_k = \mu_k N$$

Circular Motion & Gravitation

$$\Sigma F_C = ma_C = \frac{mv^2}{r}$$

$$a_C = \frac{v^2}{r}$$

$$F_G = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

Work & Energy

$$W = F_{\parallel} d = Fd \cos \theta$$

$$KE = \frac{1}{2} mv^2$$

$$PE_g = mgh$$

$$PE_e = \frac{1}{2} kx^2$$

$$W = \Delta KE$$

$$\Sigma E_0 + W_{NC} = \Sigma E$$

$$P = \frac{W}{t}$$

Name:

Date:

Period:

Scratch Work

Name:

Date:

Period:

Scratch Work

Name:

Date:

Period:

HONORS PHYSICS FALL FINAL EXAM

(Version B)



Please do not open this packet until instructed to do so!

Rules

1. **Phones, earbuds, headphones, smartwatches, computers, and other electronics must be put away in your backpacks against the wall during the entirety of the exam. If electronics are out for *any reason*, you will receive a zero and a detention!**
2. Put your name three places: On your Equation Sheet (back of this page), On your Scratch Work sheet (next page), and on the first page of the Exam Packet.
3. Bubble in the Key Version (B) on your Scantron
4. When you are finished, turn in your scantron, equation sheet, and exam packet up front.
5. Stay silent until all your classmates are finished. Electronics (especially phones, but also earbuds, headphones, laptops, smartwatches, etc.) need to be kept away until the instructor indicates that everyone is finished.

Name:

Date:

Period:

Equation Sheet

Basic Kinematics

$$\bar{v} \equiv \frac{\Delta x}{\Delta t}$$

$$\bar{a} \equiv \frac{\Delta v}{\Delta t}$$

$$\bar{v} = \frac{v + v_0}{2}$$

Kinematic Equations

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

“Old Faithful”

“Big Chalupa”

“Ain’t Got No Time”

Range Equation

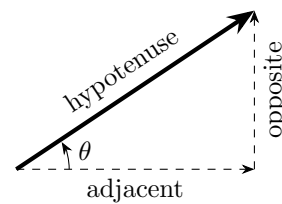
$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Vectors & Trig

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$



Forces

$$\Sigma F = ma$$

$$w = mg$$

$$f_s = \mu_s N$$

$$f_k = \mu_k N$$

Circular Motion & Gravitation

$$\Sigma F_C = ma_C = \frac{mv^2}{r}$$

$$a_C = \frac{v^2}{r}$$

$$F_G = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

Work & Energy

$$W = F_{\parallel} d = Fd \cos \theta$$

$$KE = \frac{1}{2} mv^2$$

$$PE_g = mgh$$

$$PE_e = \frac{1}{2} kx^2$$

$$W = \Delta KE$$

$$\Sigma E_0 + W_{NC} = \Sigma E$$

$$P = \frac{W}{t}$$

Name:

Date:

Period:

Scratch Work

Name:

Date:

Period:

Scratch Work

HONORS PHYSICS

FALL FINAL EXAM

(Version C)



Please do not open this packet until instructed to do so!

Rules

1. **Phones, earbuds, headphones, smartwatches, computers, and other electronics must be put away in your backpacks against the wall during the entirety of the exam. If electronics are out for *any reason*, you will receive a zero and a detention!**
2. Put your name three places: On your Equation Sheet (back of this page), On your Scratch Work sheet (next page), and on the first page of the Exam Packet.
3. Bubble in the Key Version (C) on your Scantron
4. When you are finished, turn in your scantron, equation sheet, and exam packet up front.
5. Stay silent until all your classmates are finished. Electronics (especially phones, but also earbuds, headphones, laptops, smartwatches, etc.) need to be kept away until the instructor indicates that everyone is finished.

Name:

Date:

Period:

Equation Sheet

Basic Kinematics

$$\bar{v} \equiv \frac{\Delta x}{\Delta t}$$

$$\bar{a} \equiv \frac{\Delta v}{\Delta t}$$

$$\bar{v} = \frac{v + v_0}{2}$$

Kinematic Equations

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2} at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

“Old Faithful”

“Big Chalupa”

“Ain’t Got No Time”

Range Equation

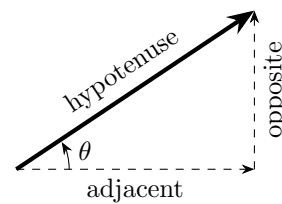
$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Vectors & Trig

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$



Forces

$$\Sigma F = ma$$

$$w = mg$$

$$f_s = \mu_s N$$

$$f_k = \mu_k N$$

Circular Motion & Gravitation

$$\Sigma F_C = ma_C = \frac{mv^2}{r}$$

$$a_C = \frac{v^2}{r}$$

$$F_G = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

Work & Energy

$$W = F_{\parallel} d = Fd \cos \theta$$

$$KE = \frac{1}{2} mv^2$$

$$PE_g = mgh$$

$$PE_e = \frac{1}{2} kx^2$$

$$W = \Delta KE$$

$$\Sigma E_0 + W_{NC} = \Sigma E$$

$$P = \frac{W}{t}$$

Name:

Date:

Period:

Scratch Work

Name:

Date:

Period:

Scratch Work

Name:

Date:

Period:

HONORS PHYSICS FALL FINAL EXAM

(Version D)



Please do not open this packet until instructed to do so!

Rules

1. **Phones, earbuds, headphones, smartwatches, computers, and other electronics must be put away in your backpacks against the wall during the entirety of the exam. If electronics are out for *any reason*, you will receive a zero and a detention!**
2. Put your name three places: On your Equation Sheet (back of this page), On your Scratch Work sheet (next page), and on the first page of the Exam Packet.
3. Bubble in the Key Version (D) on your Scantron
4. When you are finished, turn in your scantron, equation sheet, and exam packet up front.
5. Stay silent until all your classmates are finished. Electronics (especially phones, but also earbuds, headphones, laptops, smartwatches, etc.) need to be kept away until the instructor indicates that everyone is finished.

Name:

Date:

Period:

Equation Sheet

Basic Kinematics

$$\bar{v} \equiv \frac{\Delta x}{\Delta t}$$

$$\bar{a} \equiv \frac{\Delta v}{\Delta t}$$

$$\bar{v} = \frac{v + v_0}{2}$$

Kinematic Equations

$$v = v_0 + at$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

“Old Faithful”

“Big Chalupa”

“Ain’t Got No Time”

Range Equation

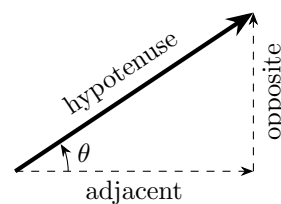
$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Vectors & Trig

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$



Forces

$$\Sigma F = ma$$

$$w = mg$$

$$f_s = \mu_s N$$

$$f_k = \mu_k N$$

Circular Motion & Gravitation

$$\Sigma F_C = ma_C = \frac{mv^2}{r}$$

$$a_C = \frac{v^2}{r}$$

$$F_G = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

Work & Energy

$$W = F_{\parallel} d = Fd \cos \theta$$

$$KE = \frac{1}{2}mv^2$$

$$PE_g = mgh$$

$$PE_e = \frac{1}{2}kx^2$$

$$W = \Delta KE$$

$$\Sigma E_0 + W_{NC} = \Sigma E$$

$$P = \frac{W}{t}$$

Name:

Date:

Period:

Scratch Work

Name:

Date:

Period:

Scratch Work